Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	UNIVERSIDAD DE VALLADOLID
Short name / acronym used in projects	UNIVERSIDAD DE VALLADOLID
Participant Identification Code (PIC), if available	999862809
Website(s)	https://innograinlab.uva.es/ https://isp.uva.es/en/
Country	Spain
Organisation type	x Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Edificio I+D, Campus Miguel Delibes, Paseo de Belén 11, 47011, Valladolid
Department(s), institute(s), research group(s) or unit(s) that would be involved	InnograinLab Research Group (GIR 464, Bioquímica y Estructura de Moléculas y Matrices Alimentarias), Institute of Sustainable Processes
Additional institutional resources relevant to collaboration	e.g. core facilities, clinical units, technology platforms, animal facilities, data infrastructures

Main contact person for this profile	Mario Martinez Martinez, Department of Agricultural Engineering
Main contact email	mmartinez@uva.es
Other contact person(s), if relevant	Name, role, email

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	x Drug delivery / formulations / nanomedicine
x Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	x Other (please specify): Analysis of structural and colloidal properties of bio-based polymers, metabolimics (profiling of bioactive compounds), advanced in vitro digestion and colonic fermentation models, extraction and processing of plant-based biopolymers.

Free Keywords for internal searchable database (please provide up to 10)	Precision nutrition, In vitro digestion, Microbiota, Metabolomics, Polymer structure, Polysacharides, Polyphenols, Glycomics, Compounds extraction, Colloidal properties
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
We are a leading group focused on designing and utilizing plant macromolecule assemblies that precisely target metabolic responses, bioactivity, and technological performance for applications in food, agriculture, pharmaceuticals, and packaging.

Why your team could be relevant to a translational eye-health and vision consortium
Our group could contribute through our expertise in designing plant-based macromolecular assemblies that modulate metabolic responses, improve bioactive delivery, and enhance material functionality. This knowledge could support the development of sustainable biomaterials and delivery systems for ocular health applications, including antioxidants and anti-inflammatory compounds relevant to retinal and eye diseases. In addition, our experience in digestion, bioaccessibility, and exposomic interactions could help investigate how nutrition, metabolism, and bioactive compounds influence eye health and disease progression. We are also experts in extraction and processing techniques, which could be harnessed to obtain and design delivery systems or bio-based molecules.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute
- Increase the precision in understanding the relationships between food structure, metabolic response, and the gut microbiome. *- Structure-function relationships in relation to building novel foods and packaging materials.* *- Analytical methods to elucidate structural features of biopolymers, including monomer composition, linkage patterns, sequencing, hydrodynamic and molecular weight, anomeric configuration and chain length distributions.* *- Chemical, enzymatic and physical modifications of plant polysaccharides and proteins.* *- Interactions between food macromolecules and small metabolites (e.g., polyphenols) in complex systems.*

- Health aspects, particularly the digestion and bioaccesibility of plant biopolymers and phenolic compounds and the role of polysaccharides as dietary fiber.
- Physical properties, including the rheological and colloidal properties of edible systems.
- The encapsulation and controlled release of active compounds for inclusion in food formulations.
- The film forming properties of plant polysaccharides and proteins with application in edible films, coatings and active packaging.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Our research infrastructure can be found in our website - https://innograinlab.uva.es/research-infrastructure/.
The group holds specialized expertise and facilities including advanced analytical equipment; molecular biology and omics resources; BSL-2 microbiology lab; static and semi-dynamic INFOGEST in vitro digestion setups; and anaerobic digesters/fermenters (2–200 L). Additionally, we have access to established agreements with specialized partners for high-throughput microbiome sequencing.
In particular, the research team laboratory locate offers two IC-PAD (850 Professional IC, Metrohm and Dionex ICS-6000, ThermoFisher Scientific), a SEC-FFF-UV-MALS-RI-IV (Agilent 1260, Wyatt Eclipse, DAWN, ViscoStar and Optilab, Agilent-Wyatt), a Fourier transformed infrared Spectrometer (ATR-FTIR Nicolet iS50, ThermoFisher), two 500 MHz solid/liquid state NMR (Agilent and Bruker), a miniSpec low field NMR (Bruker), a 6 Flex Real-Time PCR System (QuantStudio), Powder X-ray diffractometer (XRD, Bruker D8 Discover A25), an Environmental scanning electron microscope (ESEM, Quanta 200FEI), a confocal laser scanning microscope (CLSM, Leica TCS SP8X), a DLS Zetasizer Advance Pro Red Label (Malvern) and a Mastersizer (Malvern). In addition, the research team laboratory is also equipped with a UPLC-ESI-QTOF-MS (Maxis Impact, Bruker), a GC-MS (Agilent) and a MALDI-TOF-MS (Autoflex, Bruker). The research team counts with regular sample preparation and conventional microbiology labs equipped with ThermoMixer with thermoblock reaction vessels (Eppendorf F2.0); Ultraturrax homogenizer (IKA-T18), moisture analyzer (MA 150 Sartorius), Muffle (Buhler), Nitrogen-evaporator (N-EVAP, Organomation), ultrasonic water bath, incubators, laminar flow cabinets, freeze-drier, stirring plates, shaking and thermostatic water baths, centrifuges, pH meter, and several plate spectrophotometric readers. InnograinLab also count with a controlled-stress oscillatory rheometer (DHR20, TA Instruments), Differential scanning calorimeter, (DSC Q-20, TA instrument), spectrocolorimeter (Minolta CN-508i) and Texture analyzer (TA-XT2, Stable Microsystems).

Training capacity and knowledge-transfer potential

Upcoming visiting researchers would benefit from interdisciplinary training combining advanced food, biomaterial, microbiology, and analytical sciences with translational applications in health and sustainability. Training opportunities could include hands-on experience in advanced characterization techniques such as SEC-FFF-MALS, NMR, FTIR, XRD, CLSM, DLS, rheology, thermal analysis, and mass spectrometry platforms (UPLC-QTOF-MS, GC-MS, MALDI-TOF-MS), as well as molecular biology and omics-related workflows. Researchers could also receive practical training in static and semi-dynamic INFOGEST in vitro digestion models, fermentation systems and anaerobic bioreactors, microbiology under BSL-2 conditions, and microbiome-oriented studies. In addition, the group can provide expertise in plant macromolecule assemblies, food structure–function relationships, bioaccessibility and bioactivity assessment, colloidal systems, sustainable biomaterials, and data integration across physicochemical, biological, and metabolic responses. The infrastructure and multidisciplinary environment further support training in experimental design, method development, advanced analytical interpretation, and collaborative translational research.

What your team would hope to gain from participating

Access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, joint publications and longer-term strategic collaboration.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Mario Martinez Martinez	Profesor Titular	Department of Agricultural Engineering / Institute of Sustainable Processes	PTUN	A	mmartinez@uva.es
Laura Roman Rivas	Ramon & Cajal Researcher	Department of Agricultural Engineering /	Ramon & Cajal Researcher	B	lroman@uva.es

		Institute of Sustainable Processes			
Cristina Andrés Iglesias	Profesora Ayudante Doctor	Department of Agricultural Engineering / Institute of Sustainable Processes	Profesora Ayudante Doctor	C	cristina.andres.iglesias@uva.es

Main disciplines represented in your team	Biopolymer chemistry, Food chemistry, Applied biotechnology, Metabolomics, Material properties
Experience in EU / international / multi-partner collaborative projects	Mario Martínez is the Coordinator of the FutureFoods partnership co-funded by the European Union's Horizon Europe Programme under Grant Agreement 101136361 with a total funding of 1.2 M euro. Mario Martínez is the Host Principal Investigator of two MSCA-PF. Mario Martínez is the Principal Investigator of one MSCA-SE led by the University of Aveiro (Portugal). Mario Martínez coordinated the submission of two EIC Pathfinder proposals requesting 7 M euro in total that received good grades but were not funded. Laura Roman is the principal investigator of a Interreg project led by the Polytech Instituto of Braganca (Portugal).
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes x No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

1. *Bechtner, J. D., Lopez-Rodulfo, I. M., Gallego-Lobillo, P., Hosek, J., Schwab, C., & Martinez, M. M. (2025). Polyphenols non-covalently bound to artificial plant cell walls are readily accessible to fecal microbiota and transiently alter fermentation activity. Food Chemistry, 145817.*
2. *Lopez-Rodulfo, I. M., Rasmussen, M. K., Hamaker, B., & Martinez, M. M. (2026). Selective detoxification of digesta revealed how cold-pressed apple fractions modulate transepithelial glucose transport and stimulate GLP-1 secretion. Food Chemistry, 148588.*
3. *Vidal, N. P., Charlampita, M. C., Spotti, M. J., & Martinez, M. M. (2024). Multifunctional phloroglucinol-loaded pea starch coating for refrigerated salmon. Food Packaging and Shelf Life, 43, 101277.*
4. *Roman, L., Baylis, B., Klinger, K., de Jong, J., Dutcher, J. R., & Martinez, M. M. (2022). Changes to fine structure, size and mechanical modulus of phytoglycogen nanoparticles subjected to high-shear extrusion. Carbohydrate Polymers, 298, 120080.*
5. *Roman, L., Yee, J., Hayes, A. M., Hamaker, B. R., Bertoft, E., & Martinez, M. M. (2020). On the role of the internal chain length distribution of amylopectins during retrogradation: Double helix lateral aggregation and slow digestibility. Carbohydrate Polymers, 246, 116633.*

Up to 5 relevant previous projects, grants, networks or collaborations

1. *1st FutureFoodS co-funded call. Biotechnological upcycling of fruit pomace into prebiotic food ingredients (BIOPOM). Horizon Europe Framework Program and Spanish Ministry of Science and Innovation & European Union. 01/04/2026 - 01/04/2029. Amount awarded: 1,291,990.00 € (gestionado como Project Coordinator, Mario Martinez) PI: Mario Martinez (University of Valladolid). Role: Project coordinator and Principal investigator. Other PIs, Clarissa Schwab (Aarhus University), Lillian Barros (Polytechnique Institute of Braganca) and Magdalena Svanström (Chalmers University of Technology).*
2. *Marie Sklodowska Curie Actions-Postdoctoral Fellowships 2025. Decoding Ileal Metabolic Responses to Complex Dietary Structures for Precision Nutrition (PREILEAL). Horizon Europe Framework Programme (HORIZON). PIs: Ivan Misael Lopez Rodulfo & Mario Martinez. Role: co-Principal Investigator/Coordinator at host institution. Importe: 209.914 €. Ámbito europeo.*
3. *Danmarks Frie Forskningsfond - Sapere Aude 2021. 1051-00046B Non-covalent Binding to Target Prime Sites for Cardiometabolic Protection. Independent Research Fund Denmark (DFF). 01/01/2022-01/01/2026. PI: Mario Martinez. Importe: 833.511 €. Ámbito nacional.*

4. *Proyectos de Generación de Conocimiento 2024. Green crosslinking of bread waste particles for food integration and precision gut microbiome modulation (BREBIOTICS). Agencia Estatal de Investigación (AEI). 01/01/2024-31/12/2027. Amount awarded: 351.700€, PI: Mario Martinez (University of Valladolid). Role: Principal investigator.*
5. *Consolidación Investigadora 2023. Bio-inspired food packaging: harnessing plant cell wall waste infused with vacuolar polyenols (WASTEtoPACK). Spanish Ministry of Science and Innovation & European Union. 01/04/2024-01/04/2026. Amount awarded: 200.000 €. PI: Mario Martinez (University of Valladolid). Role: Principal investigator.*

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
Please indicate only what can be shared at this stage.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	x Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	x Only Researchers x Technical Staff/Innovation x Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	e.g. visa, insurance, access to facilities, language, employment status, health & safety induction

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
We are a leading group focused on designing and utilizing plant macromolecule assemblies that precisely target metabolic responses, bioactivity, and technological performance for applications in food, agriculture, pharmaceuticals, and packaging.

Capabilities or expertise you would especially like to access through the consortium
Upcoming visiting researchers would benefit from interdisciplinary training combining advanced food, biomaterial, microbiology, and analytical sciences with translational applications in health and sustainability. Training opportunities could include hands-on experience in advanced characterization techniques such as SEC-FFF-MALS, NMR, FTIR, XRD, CLSM, DLS, rheology, thermal analysis, and mass spectrometry platforms (UPLC-QTOF-MS, GC-MS, MALDI-TOF-MS), as well as molecular biology and omics-related workflows. Researchers could also receive practical training in static and semi-dynamic INFOGEST in vitro digestion models, fermentation systems and anaerobic bioreactors, microbiology under BSL-2 conditions, and microbiome-oriented studies. In addition, the group can provide expertise in plant macromolecule assemblies, food structure–function relationships, bioaccessibility and bioactivity assessment, colloidal systems, sustainable biomaterials, and data integration across physicochemical, biological, and metabolic responses. The infrastructure and multidisciplinary environment further support training in experimental design, method development, advanced analytical interpretation, and collaborative translational research.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	x Yes ☐ No
x We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☐ **Understood**

Name:

Date completed:

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)